Project Preliminary Design

In this phase, teams have the opportunity to demonstrate proficiency in the following SLOs:

- PR1. Collect a set of requirements for a simple software system
- PR3. Identify the functional and non-functional requirements in a set of requirements.
- PR4. Create a prototype of a software system to validate a set of requirements. (Building a mock-up, MVP, etc.)
- PR6. Write a requirements specification for a simple software system.
- SD1. Create a well-written, clear software architecture design.
- SD5. Create an appropriate user interface design for a software system.

During each of the 2 weeks for this phase, students have the opportunity to demonstrate proficiency in the following SLOs:

- T1. Follow effective team communication practices.
    - Contribute to the creation of the Software Requirements Specification, or, review it, and communicate feedback to the author(s).
    - Contribute to the creation of the UI Model, or, review it, and communicate feedback to the author(s).
    - Contribute to the creation of the Domain Model, or, review it, and communicate feedback to the author(s).
    - Complete the weekly status report in Teammates on time, correctly, and honestly.
- T4. Collaborate effectively in cooperative development/programming.
    - Attend the meeting, contribute, follow team contract requirements, etc. (PM: lead meeting)
- TE3. Identify configuration items and use a source code control tool in a small team-based project.
    - Create at least 1 PR this week, except the LE.
- TE5. Configure and use tools to effectively manage team communication.
    - Communicate in Microsoft Teams punctually, effectively, hospitably.
- VV13. Use a defect tracking tool to manage software defects in a small software project.
    - Uses Kanban board correctly.
- If doing coding this week:
    - SC1. Write appropriate unit tests for a small component (several functions, a single type, etc.).
    - VV12. Automate testing in a small software project.
- If not doing coding this week:
    - SC1, VV12: review the unit tests and automated tests, and give feedback to the authors in MS Teams.
- PC1: work at least 4 hours on the project in the week.

Deliver the following items:

- **Software Requirements Specification (PR1, PR3, PR6)**
    - https://www.perforce.com/blog/alm/how-write-software-requirements-specification-srs-document

        - (from the outline:) Don't worry about 3.1.3, 3.3, 4.2, 4.3, 4.4, 5.
    - Check this document in to your repo in `docs/`
    - For each user story, create a card in your GitHub Kanban board, in a User Stories / Storylog category.
    - Grades for SLOs PR1, PR3, and PR6 can be improved by revising your work after the deadline.
    - Also, consider this very thorough example: https://refactoringenglish.com/excerpts/write-an-effective-design-doc/

- **Design Models (SD1, SD5)**

    - *User Interface model*: Build a UI mockup that specifies the flow of the user's interaction with your system.
        - You can either photograph hand-drawn images of these diagrams or create them using a drawing tool (e.g., draw.io makes it relatively easy to draw and edit simple diagrams). Publish these models in your main project repo (e.g., see sample project repo).
    - *Domain Model*: Build a diagram documenting your project domain model. (https://codeburst.io/rule-your-domain-model-d4beae6806c)
    - Grades for SLOs SD1 and SD5 can be improved by revising your work after the deadline.

- **Prototype (PR4)**

    - Push an upgraded prototype to your client repo that implements at least 2 or 3 user stories. Make sure all implemented features have associated unit tests. Continue to mock the data by hardcoding it in the client app.
    - Use the Git release/tagging mechanism to mark this prototype code version as `prelim_design` (see Git Basics — Tagging).